

ARCHIT AGRAWAL

6233120435 | architagrawal000@gmail.com | www.linkedin.com/in/agrawal-archit | github.com/architagrawal | <https://agrawal-archit.vercel.app>

PROFESSIONAL EXPERIENCE

AI Software Engineer, Edplus

05/2026 - Present

- Architected and currently own a 12-agent analytics platform turning any structured dataset into interactive charts and cited answers without dataset-specific code; TypeScript, NestJS, AWS Bedrock, Step Functions.
- Designed a deterministic chart selector across 17 chart types grounded in visualization research, accepting an LLM recommendation only when it outscores the rules-based baseline.
- Quadrupled insight recall in automated reports, from 13% to 52–59% on held-out datasets, by letting an agent pick the analyses while deterministic code computes every number.
- Stopped a silent defect mislabeling 13% of rows before it reached institutional reports, reaching 100% join integrity with schema constrained LLM output and citation validation
- Benchmarked models and pipeline variants at \$5–\$89 per 100k rows, finding model choice swung F1 33x more than architecture; traced a reported 0.637 accuracy to a hand-reviewed spreadsheet and set a reproducible 0.523 baseline

Founding AI/ML Engineer, MyStage Music Inc.

07/2025 - 05/2026

- Re-architected a 14-Cloud-Function backend into one checkpointed LangGraph agent in Python on GCP Cloud Run, cutting production service dependencies 64%.
- Built a Playwright ingestion pipeline handling 70,000+ records a day from 650+ venues, with transactional locking against duplicate work and Gemini entity resolution raising dataset accuracy 25%.
- Delivered sub-50 ms search on FastAPI and Algolia with real-time Firestore sync, covered by 200+ pytest tests and Logfire tracing.

Software Developer, Edplus

09/2023 - 05/2025

- Led end-to-end development of a multi-tenant RAG assistant, React front end to retrieval layer, used by 1,000+ faculty to author courses reaching 60,000+ students.
- Reduced transcript-analysis time 16x, from four hours to 15 minutes, with a Neo4j knowledge graph queried by schema-validated LLM generated Cypher.
- Enforced per-college data isolation at the retrieval layer so one tenant's material cannot surface in another's, making onboarding a config change, not a deployment
- Built a Prompt Flow evaluation harness scoring grounded, relevance and coherence, so every prompt change shipped on evidence rather than opinion

Student Researcher, Arizona State University

08/2024 - 05/2025

- Built a PyTorch and Diffusers benchmark for text-to-video models scoring physical plausibility separately from visual quality, with seed, resolution and sampler held fixed.
- Demonstrated weak correlation between visual quality and physical plausibility: the model ranked first in human preference but last in causal consistency.

Data Research Aide, Knowledge Exchange for Resilience

05/2024 - 08/2024

- Automated **high-volume ETL workflows using Python and PostgreSQL** processing 10,000+ daily faculty records with validation layers improving downstream data reliability
- Cut p95 retrieval latency 60% at 95% recall with a hybrid FAISS and Neo4j search layer.
- Built a FastAPI and pgvector recommendation service reaching 85% top-k precision for collaborator matching, fed by an ETL ingesting 10,000+ profiles a day, alongside 20+ REST APIs on .NET 8.

Software Engineer, Zeus Learning

01/2022 - 08/2023

- Built a Redis-backed desk-demand forecasting service used across 300+ offices at Fortune 500 companies, with reported desk occupancy up 30%.
- Cut deploy time 70% and production incidents 40% by containerizing the release pipeline and adding SonarQube quality.
- Migrated a C# and .NET monolith to Dockerized Kubernetes microservices incrementally behind the existing API, cutting infrastructure cost 20% and footprint 35%

Software Developer Intern, EAT.FIT

09/2021 - 12/2021

- Cut driver-tracking server cost 45% with distance-based polling and client-side interpolation that kept map movement smooth.

TECHNICAL SKILLS

Languages: Python, Go, C#, JavaScript, TypeScript, SQL.

LLM & Agentic AI: Hugging Face (Transformers, Diffusers), LangGraph, CrewAI, Pydantic AI, MCP, LangChain, LlamaIndex, RAG Pipelines, Vector Stores (Pinecone/FAISS), OpenAI API, OCR (Tesseract/PaddleOCR), Voice Generation (Bark), Video Generation (SVD).

HPC & Compute: CUDA, C++, Linux (Bash/Shell), GPU Profiling (Nsight), Docker, Kubernetes, AWS (EC2, Lambda), GCP (Vertex AI).

Web & Backend: FastAPI, Django, Flask, .NET Core, Node.js, React.js, GraphQL.

Data & Observability: SQLAlchemy, PyTorch, TensorFlow, NumPy, Pandas, Logfire (LLM Tracing), Locust, Redis, RabbitMQ.

Databases: PostgreSQL, MySQL, Redis, Neo4j (Graph DB), Pinecone (Vector DB), DynamoDB

Cloud & DevOps: AWS (EC2, Lambda, EKS, SQS, S3), Google Cloud Platform, Docker, Kubernetes, Jenkins, GitLab, Nginx, RabbitMQ

EDUCATION

Master of Science, Computer Science

08/2023 - 05/2025

Arizona State University (ASU)

GPA: 4.0

Bachelor of Technology, Information and Communication Technology

08/2018 - 05/2022

Dhirubhai Ambani Institute of Information and Communication Technology